

DEFORMING FORCES

When a force acts on an object that cannot move, or when a number of different forces act in different directions, the whole object changes shape. The type of distortion an object undergoes depends on the number, directions, and strengths of the forces acting upon it, and on its structure and composition—if it is elastic (returns to its original shape) or plastic (deforms easily but does not return to its original shape). Brittle materials fracture, creep, or show fatigue if forces are applied to them.



Compression

When two or more forces act in opposite directions and meet in an object, it compresses and bulges.



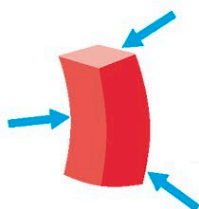
Tension

When two or more forces act in opposite directions and pull away from an elastic object, it stretches.



Torsion

Turning forces, or torques, that act in opposite directions twist the object.



Bending

When several forces act on an object in different places, the object bends (if malleable) or snaps.

Resultant forces

A force is balanced when another force of the same strength is acting in the opposite direction. Overall, this has the same effect as no force at all.



RESULTANT FORCE: 0 N

When opposing teams pull with equal force, the resultant force is 0 N.



RESULTANT FORCE: 100 N

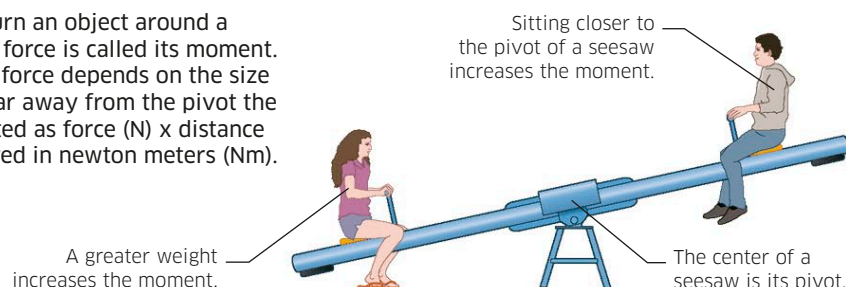
One team pulls with more force than the other. The resultant force is 100 N.

TURNING FORCES

Instead of just moving or accelerating an object in a line, or sending an object off in a straight line in a different direction, forces can also be used to turn an object around a point known as an axis or a pivot. This kind of force works on wheels, seesaws, and fairground rides such as carousels. The principles behind these turning forces are also used in simple machines (see pp.106–107).

Moment

When a force acts to turn an object around a pivot, the effect of the force is called its moment. The turning effect of a force depends on the size of the force and how far away from the pivot the force is acting. Calculated as force (N) x distance (m), moment is measured in newton meters (Nm).



Centripetal forces

A constant force has to be applied to keep an object turning in a circle, obeying Newton's first law of motion (see pp.98–99). Known as centripetal force, it pulls the turning object toward the center of rotation—imagine a yo-yo revolving in a circle on its string—continually changing its direction, while the motion changes its speed. Without this force, the object would move in a straight line away from the center.

